

Dhruv Maheshwari

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EDUCATION

PES University, Bengaluru

2027

Bachelor of Technology - **Computer Science and Engineering**

Prof. CNR Rao **Scholarship** (40% Tuition Merit Award), 4× **Distinction Award**

SKILLS

Languages: Python, SQL (MySQL), JavaScript

Frameworks & Libraries: FastAPI, Uvicorn, Pydantic, Pandas, React (Vite), FastEmbed, LangChain, LangGraph, ChromaDB, Three.js

Tools & Platforms: Git, GitHub, Docker, Kubernetes, AWS Bedrock, Agno, Vercel, Azure Dev Console, HuggingFace, Blender

Concepts: REST API design, Multi-agent AI systems, Retrieval Augmented Generation (RAG), LLMs, LLM inference, Embeddings, CI/CD, Prompt Engineering

Coursework: Data Structures and Algorithms, Operating Systems, Big Data, Software Engineering, Computer Networks, Cloud Computing, Object Oriented Analysis and Design, Digital Twin

EXPERIENCE

Intern – Technology

Jun – Aug 2026

AceNet Consulting

Python · AWS Bedrock · FastAPI · Uvicorn · Azure Dev Console · LangChain

- Built the backend for **CFO Copilot**, a **multi-agent LLM platform** using **Python, FastAPI, and AWS Bedrock**, enabling executives to query macroeconomic, industry, and financial data through **3 specialist agents** over **24 analysis modes**.
- Designed an **LLM query-routing and orchestration layer** with the **AWS Bedrock API**, parallel execution, and **20+ REST endpoints** across chat, agent reasoning, scenario analysis, source lookup, and more.
- Engineered **ingestion and analytics pipelines** over **100K+ government-sourced Vahan records** and real data feeds; implemented **source-aware confidence scoring and verification** that traces figures to source data.

AI App Development Intern

Jun – Jul 2025

xForm Solutions

Python · Whisper · NLLB-200 · Sentence-BERT

- Built a **multilingual speech-understanding pipeline** for code-switched Hindi, Tamil, and English audio, chaining Whisper ASR, NLLB-200 translation, and Sentence-BERT intent classification to serve Tier 2/3 city users.
- Benchmarked ASR** on 150 Google FLEURS clips across English, Hindi, and Tamil, reaching **6.4% CER** on English, and added audio resampling, ASR error correction, and an **evaluation harness** for repeatable benchmarking.

PROJECTS

pRAGma - Full-Stack RAG Application

Python · FastAPI · React (Vite) · Prompt Engineering · FastEmbed · Groq API · ChromaDB

github.com/dhruvm-04/pRAGma

- Built a **full-stack RAG application** that ingests PDFs as **asynchronous background jobs** with live progress, splits them into overlapping **1,200-character chunks**, embeds them locally with `bge-small-en-v1.5`, and serves persistent vector search behind **4 FastAPI endpoints**.
- Implemented **5 retrieval modes** (direct, multi-query, decomposition, step-back, HyDE) with grounded prompts that decline on insufficient context; every answer exposes **retrieved chunks**, similarity scores, per-stage latency, tokens, and cost.

MusiConvert - Secure Peer-to-Peer Playlist Sharing Platform

Python · Socket Programming · Fernet Encryption · Multithreading · WebSockets

github.com/dhruvm-04/MusiConvert

- Implemented **Selective Repeat ARQ** from scratch over **raw UDP** (sliding window, per-packet ACKs, retransmission timers) to carry **Fernet-encrypted** playlists between peers, with multithreaded socket handling for concurrent transfers.
- Verified in-order, **lossless delivery** of **200-packet** streams at up to **40% simulated packet loss**, and built a **fuzzy track matcher** that converts playlists between Spotify and YouTube Music despite inconsistent titles and metadata.

aiCFO - Multi Agent Advisory Application

Python · LangChain · LangGraph · Groq · Pandas · yFinance · Pydantic github.com/dhruvm-04/aiCFO

- Orchestrated a **parallel multi-agent workflow** in LangGraph: Automotive and Macro agents run concurrently over shared state before a CFO agent synthesizes, with **17 LLM-callable tools** over Vahan registrations (2022–2026) and macro series.
- Enforced **strict Pydantic schemas** on every agent so outputs arrive as typed findings, trends, caveats, and recommendations, and bounded the tool-calling loop to keep runs predictable.

paraTile - Distributed Kafka-Based Image Processing Platform

Python · FastAPI · Apache Kafka · Pillow · Multithreading · Distributed Systems github.com/dhruvm-04/paraTile

- Engineered a **distributed master-worker image-processing pipeline** using **12 Kafka task partitions** and **3 Kafka topics** for task distribution, result collection, and worker monitoring.
- Implemented **multithreaded worker orchestration** with **3-second heartbeats**, **10-second liveness detection**, and in-memory retention of the **20 most recent jobs**.

Social Media Platform - MySQL

Python · MySQL · PyMySQL github.com/dhruvm-04/SocialMediaPlatform

- Built a **full-stack social platform** on a **relational MySQL schema** with registration and login, user profiles, post CRUD, like toggles, comments with author and post-owner delete permissions, **friend requests and acceptance**, and an admin panel.

ACHIEVEMENTS & LEADERSHIP

- **Won 1st Place at the PES University Digital Twin & XR Hackathon 2026.**
- Taught **3D modeling** and **game-ready asset creation** in Blender to **~50 students**, and served as **Head of Design** for **Maaya 2025**, PES University’s techno-cultural fest.
- Served as **Under Secretary General** of PES University’s Model United Nations society, **MUNSoc PESUECC**, and **led chassis design and fabrication** for the automotive club **Team Vegavath** using Blender and Fusion 360.

CERTIFICATIONS

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|--|------------------------------|
| • AWS Prompt Engineering Foundation Prompt Engineering · Responsible AI | Link
2025 |
| • IBM Fundamentals of AI Machine Learning · Ethical AI | 2025 |
| • Google IT Automation with Python OOPs · Python · DSA | 2023 |